

CSCI 160 Homework 1 — Answer Key

1. Add the binary numbers $10101 + 11011 + 111$.

Add the first two, then add the third:

$$\begin{array}{r} 10101 \\ 11011 \\ \hline 110000 \end{array} \qquad \begin{array}{r} 110000 \\ + 111 \\ \hline 110111 \end{array}$$

Answer: $10101 + 11011 + 111 = \mathbf{110111}_2$ (check: $21 + 27 + 7 = 55_{10} = 110111_2$).

2. Express each number in decimal.

(a) $(11010.101)_2 = 16 + 8 + 2 + \frac{1}{2} + \frac{1}{8} = \mathbf{26.625}$

(b) $(4021.2)_5 = 4(125) + 0(25) + 2(5) + 1 + \frac{2}{5} = 500 + 10 + 1 + 0.4 = \mathbf{511.4}$

(c) $(127.4)_8 = 1(64) + 2(8) + 7 + \frac{4}{8} = 64 + 16 + 7 + 0.5 = \mathbf{87.5}$

(d) $(B65F)_{16} = 11(4096) + 6(256) + 5(16) + 15 = 45056 + 1536 + 80 + 15 = \mathbf{46687}$

3. Convert decimal 247 to binary (repeated division by 2, read remainders bottom-up):

$$\begin{array}{rcl} 247 \div 2 & = & 123 \text{ r } 1 \\ 123 \div 2 & = & 61 \text{ r } 1 \\ 61 \div 2 & = & 30 \text{ r } 1 \\ 30 \div 2 & = & 15 \text{ r } 0 \\ 15 \div 2 & = & 7 \text{ r } 1 \\ 7 \div 2 & = & 3 \text{ r } 1 \\ 3 \div 2 & = & 1 \text{ r } 1 \\ 1 \div 2 & = & 0 \text{ r } 1 \end{array}$$

Answer: $247_{10} = \mathbf{11110111}_2$.

4. Convert decimal 153 to octal (repeated division by 8):

$$\begin{array}{rcl} 153 \div 8 & = & 19 \text{ r } 1 \\ 19 \div 8 & = & 2 \text{ r } 3 \\ 2 \div 8 & = & 0 \text{ r } 2 \end{array}$$

Answer: $153_{10} = \mathbf{231}_8$ (check: $2(64) + 3(8) + 1 = 153$).

5. Multiply 10110×101 .

$$\begin{array}{r} 10110 \\ \times 101 \\ \hline 10110 \\ 00000 \\ 10110 \\ \hline 1101110 \end{array}$$

Answer: $10110 \times 101 = \mathbf{1101110}_2$ (check: $22 \times 5 = 110_{10} = 1101110_2$).

6. Conversions.

(a) $(10110011.1011)_2$.

Octal (group by 3 from the point): $\underbrace{010}_2 \underbrace{110}_6 \underbrace{011}_3 \cdot \underbrace{101}_5 \underbrace{100}_4 = \mathbf{263.54_8}$.

Hex (group by 4 from the point): $\underbrace{1011}_B \underbrace{0011}_3 \cdot \underbrace{1011}_B = \mathbf{B3.B_{16}}$.

(b) $(4F2A)_{16} = \underbrace{0100}_4 \underbrace{1111}_F \underbrace{0010}_2 \underbrace{1010}_A = \mathbf{0100\ 1111\ 0010\ 1010_2}$.

(c) $(572.6)_8 = \underbrace{101}_5 \underbrace{111}_7 \underbrace{010}_2 \cdot \underbrace{110}_6 = \mathbf{101111010.110_2}$.

7. Complements.

(a) 10110100 : 1's complement = **01001011**; 2's complement = $01001011 + 1 = \mathbf{01001100}$.

(b) 246700 : 9's complement = **753299**; 10's complement = $753299 + 1 = \mathbf{753300}$.

(c) $C3DF$: 15's complement = $3C20$, so the 16's complement = $3C20 + 1 = \mathbf{3C21}$.

8. Subtraction by complements.

(a) 2's complement: $1010100 - 1000011$.

2's comp. of $1000011 = 0111101$.

$$\begin{array}{r} 1010100 \\ + 0111101 \\ \hline 1\ 0010001 \end{array}$$

End carry = 1 $\Rightarrow$ result is positive; discard it. **Answer: $0010001_2 = 17_{10}$.**

(b) 10's complement: $72532 - 3250$ (align to 5 digits: 03250).

10's comp. of $03250 = 96750$.

$$\begin{array}{r} 72532 \\ + 96750 \\ \hline 1\ 69282 \end{array}$$

Discard the end carry. **Answer: 69282 .**

(c) 1's complement: $1100100 - 1010011$.

1's comp. of $1010011 = 0101100$.

$$\begin{array}{r} 1100100 \\ + 0101100 \\ \hline 1\ 0010000 \end{array}$$

End-around carry: add the carry back in, $0010000 + 1 = 0010001$. **Answer: $0010001_2 = 17_{10}$.**

9. Add $791 + 658$ in BCD.

$$\begin{array}{r} 0111\ 1001\ 0001 \\ +\ 0110\ 0101\ 1000 \\ \hline \end{array}$$

- Units: $0001 + 1000 = 1001$ ($= 9 \leq 9$, no correction).

- Tens: $1001 + 0101 = 1110$ ($= 14 > 9$, add 0110): $1110 + 0110 = 1\ 0100 \rightarrow 4$, carry 1.
- Hundreds: $0111 + 0110 + 1 = 1110$ (> 9 , add 0110): $\rightarrow 0100 = 4$, carry 1.
- Thousands: carry $\rightarrow 1$.

Answer: 0001 0100 0100 1001_{BCD} = **1449**.

10. Signed 2's-complement, 8 bits.

$$+29 = \mathbf{00011101}, \quad +49 = \mathbf{00110001}.$$

-49 is the 2's complement of 00110001: flip $\rightarrow 11001110$, add 1 $\rightarrow \mathbf{11001111}$.

$$\begin{array}{r} 00011101 \\ + 11001111 \\ \hline 11101100 \end{array}$$

(No carry out is kept.) The result 11101100 has sign bit 1, so it is negative; its magnitude is the 2's complement: $00010100 = 20$.

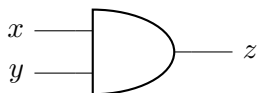
Answer: $(+29) + (-49) = 11101100_2 = -\mathbf{20}_{10}$.

11. Binary logic.

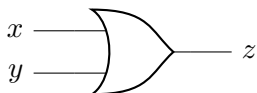
(a) Truth tables.

x	y	$x \cdot y$	x	y	$x + y$	x	x'
0	0	0	0	0	0	0	1
0	1	0	0	1	1	1	0
1	0	0	1	0	1		
1	1	1	1	1	1		

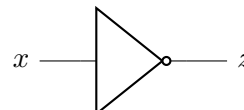
(b) Logic gates (inputs labeled, output z).



AND: $z = x \cdot y$



OR: $z = x + y$



NOT: $z = x'$